PROJECT OVERVIEW

VirtualGlove Project Overview

Architecture, controls, security, deployment, and project status.

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Move your hand. Play the game.

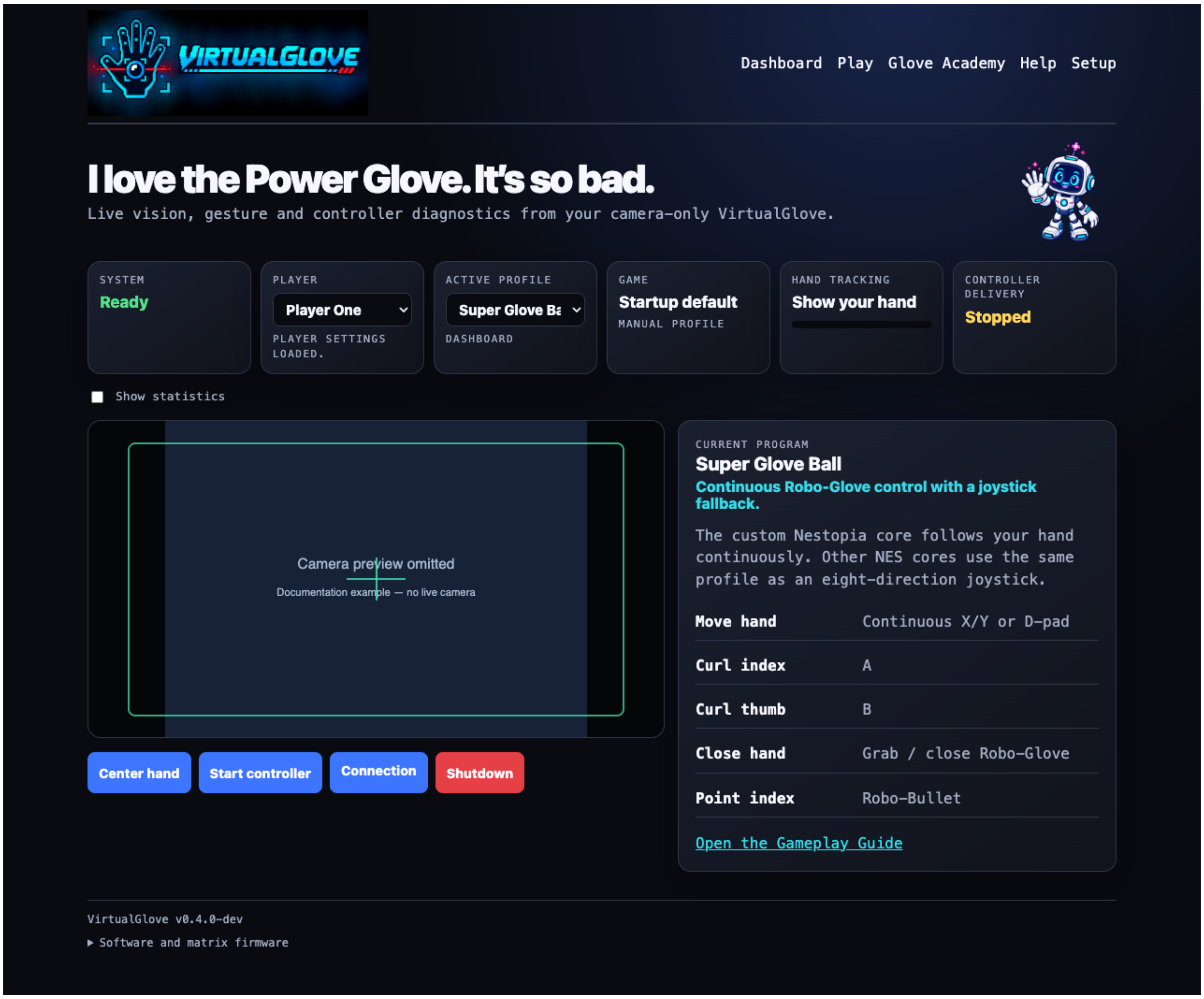
VirtualGlove turns hand movement and gestures into responsive RetroPie controls using an ordinary USB camera and a **VirtualGlove Controller** built on the Arduino UNO Q. Wear a plain glove or use your bare hand-there are no sensors, wires, or electronics to add to it.

Move to steer. Curl fingers for buttons. Roll, push, pull, grab, throw, and punch. VirtualGlove recognizes the pose, sends authenticated controller input across your local network, and lets RetroArch see a virtual gamepad or a native Power Glove controller.

Current project version: 0.4.0 · Public candidate: [v0.4.0-rc.7]

Why VirtualGlove?

- **Camera-only play:** use a standard UVC camera rather than modifying a glove.
- **Two styles of NES control:** ordinary joystick output through FCEUmm and continuous native movement for Super Glove Ball through `lr-nestopia-powerglove`.
- **Fast, direct tracking:** MediaPipe Hands processes the newest camera frame and sends the latest valid hand coordinate without a settling tail.
- **Family-friendly learning:** Pixel Pal guides players through 16 Glove Academy lessons without sending accidental input to a game.
- **Personal setup:** each player can save a hand centre, movement reach, joystick centre box, gesture sensitivity, and Academy progress.
- **Game-aware and safe:** registered games select their profile automatically; stale, lost, or unauthenticated input returns to neutral.
- **Local by design:** video stays on the Controller. Pairing and controller traffic are authenticated between your own devices.



VirtualGlove Dashboard with camera and controller status

What you need

- An Arduino UNO Q provisioned through Arduino App Lab
- A Raspberry Pi with a working RetroPie installation
- A UVC-compatible USB camera and powered USB hub
- A physical controller for RetroArch setup and recovery
- Both devices on the same trusted local network with internet access during installation
- Your own legally obtained games-VirtualGlove includes no ROMs or BIOS files

New to the hardware? Start with [Build your own](#) for the parts, expected cost, and difficulty.

Install VirtualGlove

These steps install the current public release candidate. Install the **same version on both devices** and close any running RetroArch game first. The scripts verify their downloads, ask for administrator access when needed, and preserve existing pairing and player settings during an update.

1. Prepare the Controller

Finish the UNO Q's App Lab setup, connect it to your network, and attach the camera through the powered hub. The camera may also be connected after installation.

2. Install the Controller software

Open a terminal on the UNO Q and run:

```sh
SH
cd /home/arduino
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/download/v0.4.0-rc.7/install-uno-q.sh
bash install-uno-q.sh --development v0.4.0-rc.7
```

On a first installation, the installer suggests **virtualglove** as the Controller name, making its usual address `virtualglove.local`. Press Enter to accept it or type a different family-friendly name. Updates preserve the existing name.

3. Check the Controller

Open the Dashboard address printed by the installer-normally `http://virtualglove.local:8088/dashboard`. The installer configures automatic startup, the matrix display, guarded camera recovery, and its required host helpers.

4. Install the RetroPie software

Open a terminal on the Raspberry Pi and run:

```sh
SH
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/download/v0.4.0-rc.7/install-retropie.sh
bash install-retropie.sh --development v0.4.0-rc.7
```

If an older Buster-based RetroPie reports that its Raspbian repository has no Release file, stop and follow Buster package source moved, then rerun this step. The installer does not silently rewrite operating-system repositories.

5. Pair the devices

Open the secure Setup address printed by the installer-normally `https://virtualglove.local:8443/setup`-save the RetroPie address, and choose **Pair with RetroPie**. The guided one-time-code method is recommended. After confirming the browser certificate against the physical Matrix ID, the optional **Trust this Controller** step removes future privacy warnings on that phone or computer.

6. Set up a player

Choose a player, position the camera, and use **Center hand**. Open **Glove Academy** to learn the gestures and adjust movement reach or sensitivity only if needed.

7. Play

Select **Start controller**, launch a registered game, and confirm the expected profile. FCEUmm uses joystick mode; Super Glove Ball can also use the optional native core selected from RetroPie's per-ROM launch menu.

The complete Installation Guide has first-install checkpoints, illustrated pairing, camera advice, native-core setup, updates, backups, and troubleshooting. Use it as the authoritative setup reference.

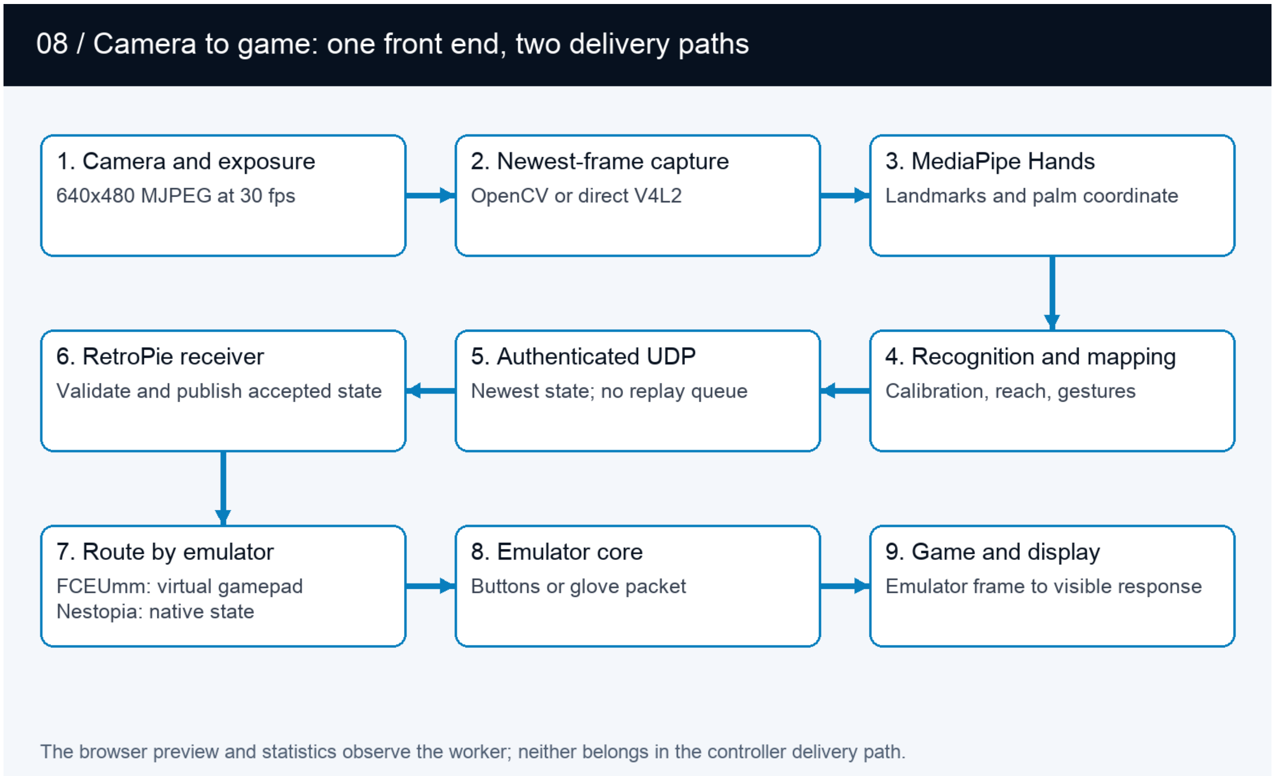
What can you play?

VirtualGlove includes nine reusable Programs A-I plus dedicated mappings for Bad Street Brawler and Super Glove Ball. The same recognition settings follow the player across games; profiles change only what the recognized movements and gestures send to the console.

Path	What it provides
FCEUmm	A nine-region joystick layout: centre stop, four directions, four diagonals, plus mapped A/B and special gestures.
Super Glove Ball with FCEUmm	A complete joystick-mode fallback that can always be selected for testing or play.
Super Glove Ball with `lr-nestopia-powerglove`	Continuous native X/Y and Z, Start, grab/catch, release/throw, Robo-Bullet fire, and Power Punch.

The Gameplay Guide shows every gesture, Program, game mapping, objective, and practice challenge. The Native Emulation guide explains why the two emulator paths feel different.

How it works



End-to-end VirtualGlove flow from camera to game

1. The camera delivers its newest frame to the VirtualGlove Controller.
2. MediaPipe Hands finds the palm, wrist, and finger landmarks.
3. Shared recognition turns those landmarks into position, fingers, rolls, depth motion, and menu poses.
4. The Controller sends the newest authenticated state to RetroPie.
5. RetroPie publishes either virtual gamepad input or native glove state for the selected emulator core.

Pairing is tied to a private shared key rather than one permanent IP address. If DHCP changes an address or `.local` resolution temporarily fails, the paired devices can rediscover one another on the local network without broadcasting controller states or requiring a new pairing.

Meet Pixel Pal

Pixel Pal helps players learn, personalize, test, and troubleshoot without turning setup into an engineering exercise.

- **Glove Academy** teaches all 16 movements and gestures while game output is paused.
- **Tune gestures** records guided examples and previews a conservative adjustment before saving it.
- **Find the best camera settings** compares only choices supported by the attached camera and changes nothing until the player accepts a recommendation.
- **Rock Paper Scissors** provides a camera-controlled practice game that does not require RetroPie.

Controller pages

Page	Purpose
Dashboard · `/dashboard`	See the camera, selected game profile, tracking state, and generated controls.
Play · `/play`	Challenge Pixel Pal to Rock Paper Scissors.
Glove Academy · `/learn`	Learn gestures, set movement reach, and personalize recognition safely.
Setup · `/setup`	Manage players, camera choices, console pairing, games, backups, and display preferences.
Help · `/help`	Read the complete manuals and printable PDFs directly on the Controller.

Documentation

Start here

You want to...	Read...
Install, pair, and play your first game	Installation Guide
Learn gestures, Programs, and game controls	Gameplay Guide
Choose or troubleshoot a camera	Camera Guide
Recognize matrix animations and messages	Matrix Display Guide
Find a quick command or status reminder	Quick Reference
Solve a problem by symptom	Troubleshooting

Go deeper

You want to...	Read...
See the current components and data flow	Architecture
Understand joystick and native emulation	Native Emulation Explained
Review proven Super Glove Ball behavior	Native Compatibility Record
Look up every setting and command	Configuration Reference
Follow the one-week engineering process and experiments	Engineering Journey
Review security and pairing boundaries	Security Policy
Check dependencies and third-party terms	Third-party Notices
Contribute code or documentation	Contributing Guide

Printable editions of all maintained guides are available in `output/pdf/`. The Controller serves the same documentation from its local Help page.

Project status

Version 0.4.0 freezes the proven CPU MediaPipe Hands path after extensive camera, tracking, reacquisition, network, emulator, and gameplay testing. The current candidate is ready for people who want to try VirtualGlove on their own UNO Q and RetroPie hardware. Different cameras, rooms, players, and Raspberry Pi installations remain valuable real-world tests.

Use Setup's **Download system report** when asking for help. It records useful software, camera, controller, and connection health without including video, pairing keys, player calibration, ROM names, or network addresses.

Contributing and licensing

Issues, careful test reports, documentation improvements, and code contributions are welcome. Please read the Contributing Guide before opening a change. Release history is kept in the Changelog.

VirtualGlove is maintained by **Iain Bennett** and is licensed under the MIT License. The modified Nestopia core is GPLv2 software and remains separate from the MIT application. Nintendo, NES, Power Glove, and the named games belong to their respective owners. See Third-party Notices for dependency, model, asset, and native-core licensing details.